

Real-World Applications of Parallel Computing and Networked Systems

① Parallel Computing Application: Weather Forecasting

→ Use Case: National meteorological centers use high-performance computing clusters to simulate weather patterns.

→ How parallel computing is used?

- The Earth's atmosphere is divided into a 3D grid.
- Mathematical models calculate temperature, pressure, humidity, wind speed at each grid point.

→ Why it's important?

- Enables risk forecasting for disaster management.
- High-resolution predictions save lives and aid agriculture, aviation and logistics.

② Networked Systems - Application: Online Banking.

→ Use Case: Banks like SBI, HDFC, on etc etc operate networked systems to allow real-time access to accounts, ATMs and online services.

→ How Networked Systems Are used?

- Customer data is stored in centralized databases accessed via secure networks.
- ~~Network~~ Load balancers and redundant servers ensure 24/7 availability.

→ Why it's important?

- Enables seamless transactions from anywhere (RTGS, UPI, NEFT).
- Ensure security, Scalability and business continuity.
- Supports millions of users with high availability and speed.